

Tube delivers oxygen from the cylinder to the diver's mouthpiece.

Divers wearing modern scuba gear

Oxygen cylinder

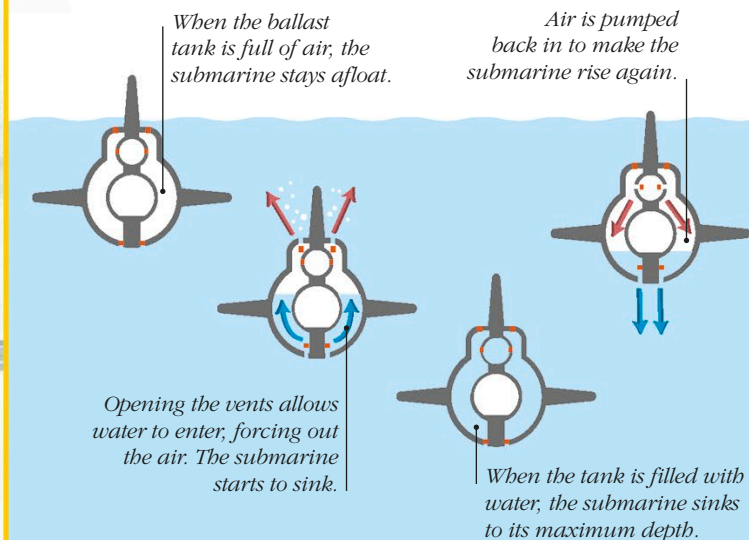
BREATHING UNDERWATER

Although the Italian inventor Leonardo da Vinci designed a diving suit about 500 years ago, it was not until 1943 that the first practical diving system became a reality. The Frenchmen Jacques Cousteau and Émile Gagnan invented the self-contained underwater breathing apparatus (SCUBA), which allowed divers to swim underwater while breathing oxygen from cylinders strapped to their backs.

A small platform called a bridge is used for observation when the submarine has surfaced.

Submarine on a naval exercise in Hawaii

DIVING AND RISING



All submarines have a double skin. In between the inner skin and the outer skin are large spaces, called ballast tanks, that can hold water. To dive, the submarine pumps water in to flood these tanks, making the vessel heavy enough to sink. To resurface, the submarine pumps the water out to make it lighter.

SUBMARINES TODAY

Built for stealth, most modern submarines are used as machines of war. Navies today have both diesel-electric (below) as well as nuclear-powered submarines. The nuclear-powered vessels can stay at sea for months without needing to resurface, and can travel around the world without needing to refuel. The nuclear reactors driving these submarines produce an unlimited amount of power without the need for oxygen. The only reasons a submarine may need to return to the surface are to replenish food and to remove waste.